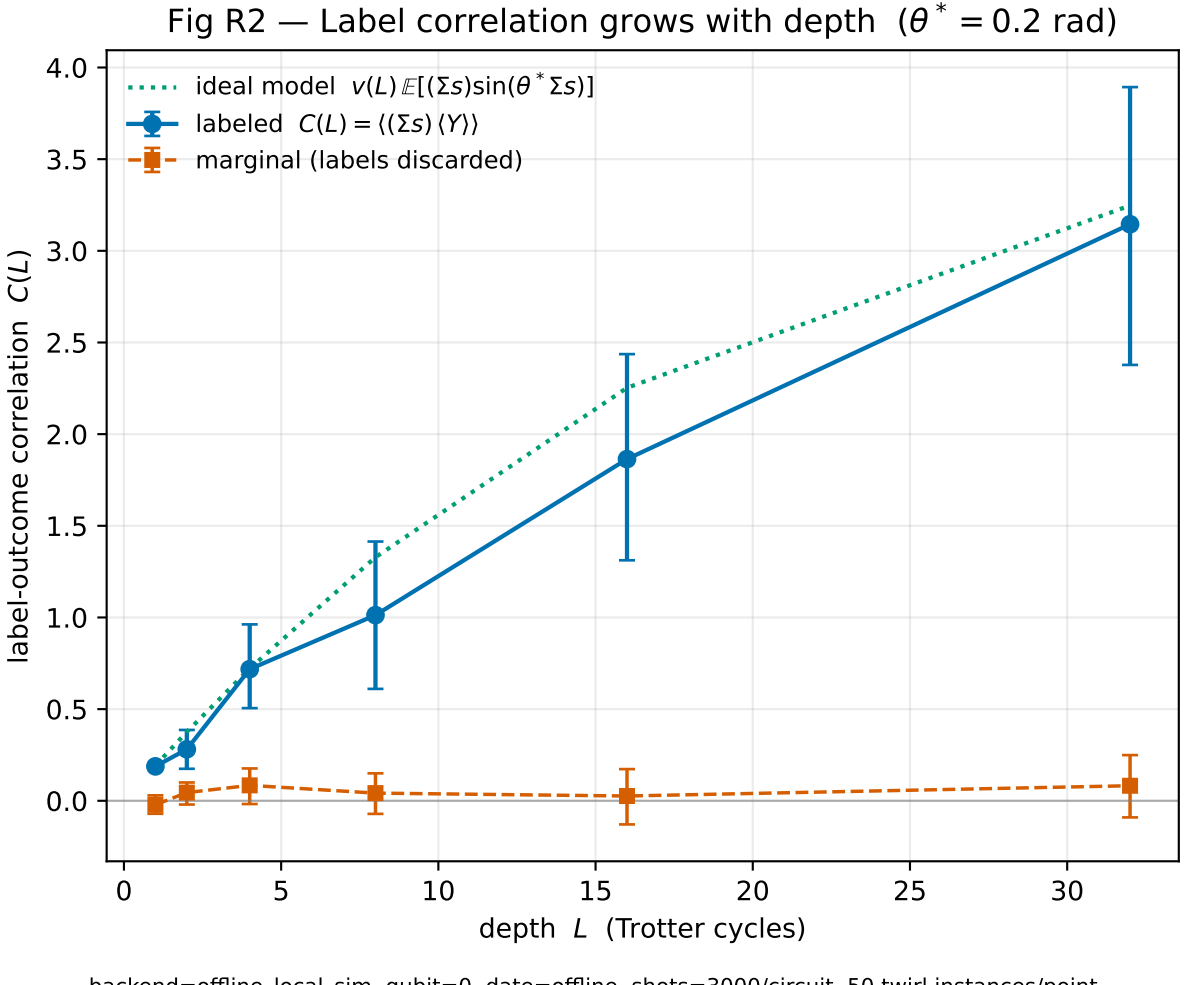
Fig R2 — Label correlation grows with depth ($\theta^* = 0.2$ rad)
ideal model $v(L)\,\mathbb{E}[(\Sigma s)\sin(\theta^*\Sigma s)]$
labeled $C(L) = \langle(\Sigma s)\langle Y\rangle\rangle$
marginal (labels discarded)
label-outcome correlation $C(L)$
depth L (Trotter cycles)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
25
30
backend=offline_local_sim, qubit=0, date=offline, shots=3000/circuit, 50 twirl instances/point